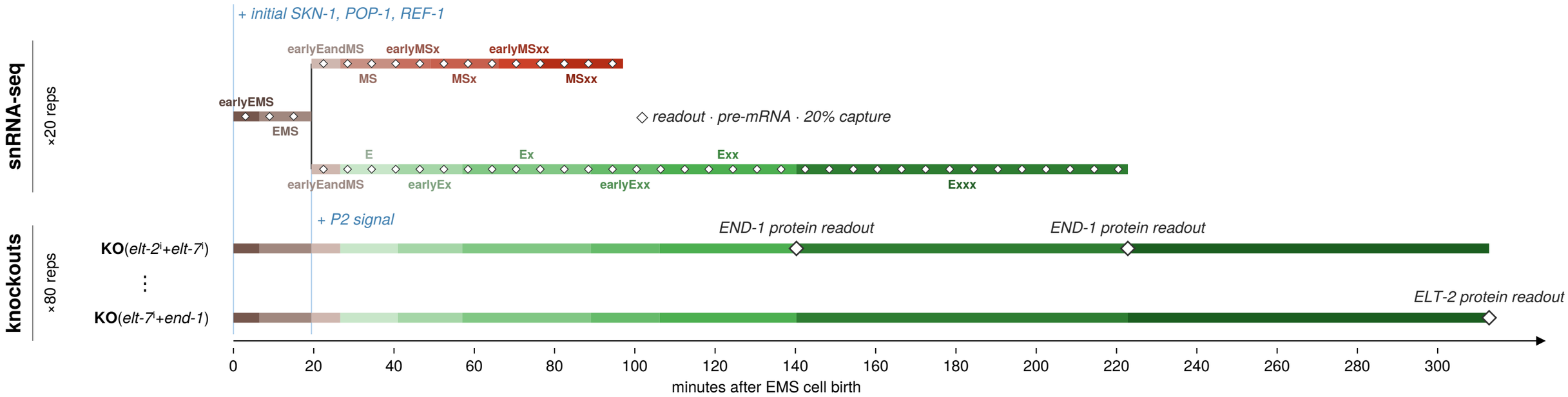
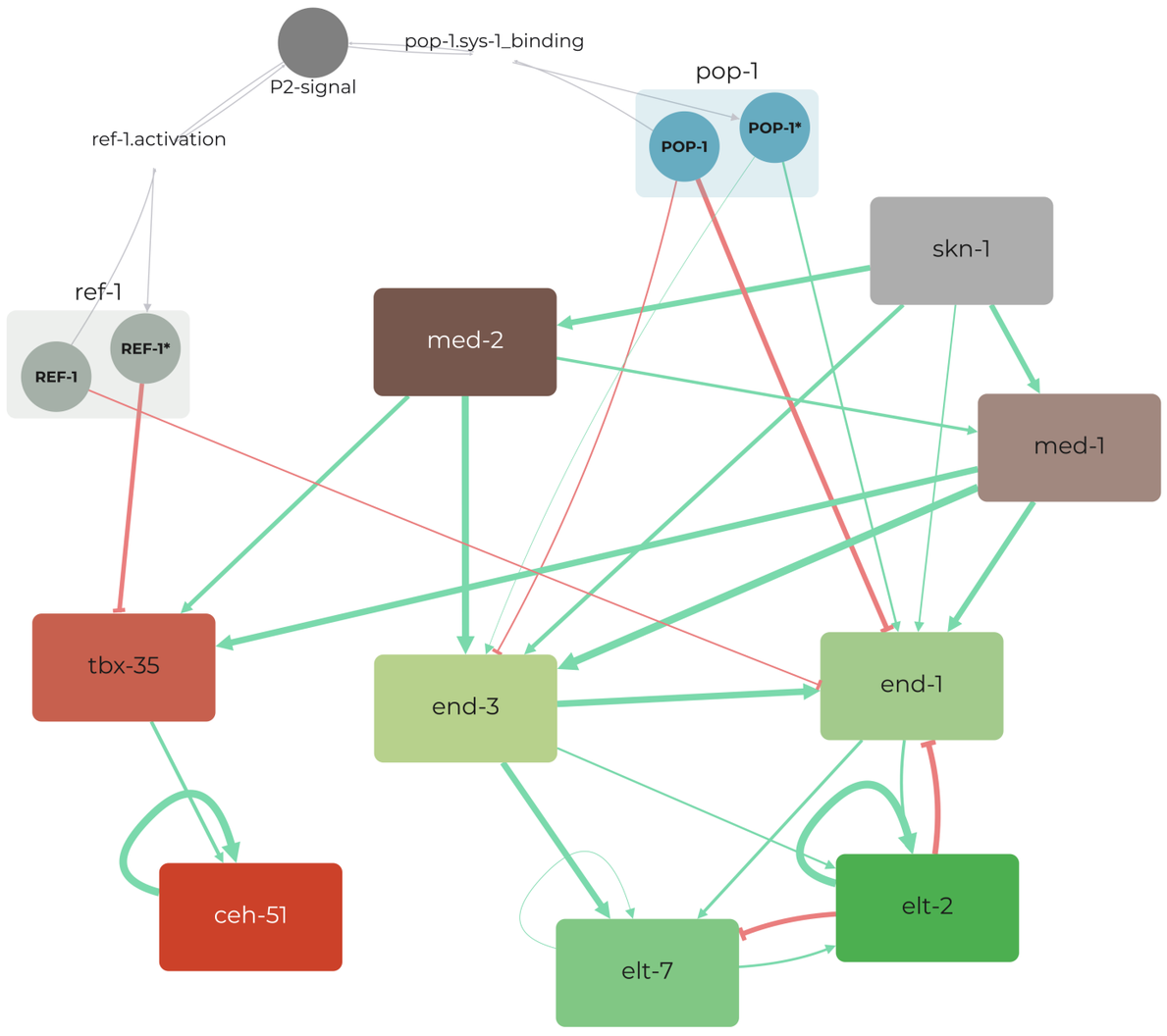
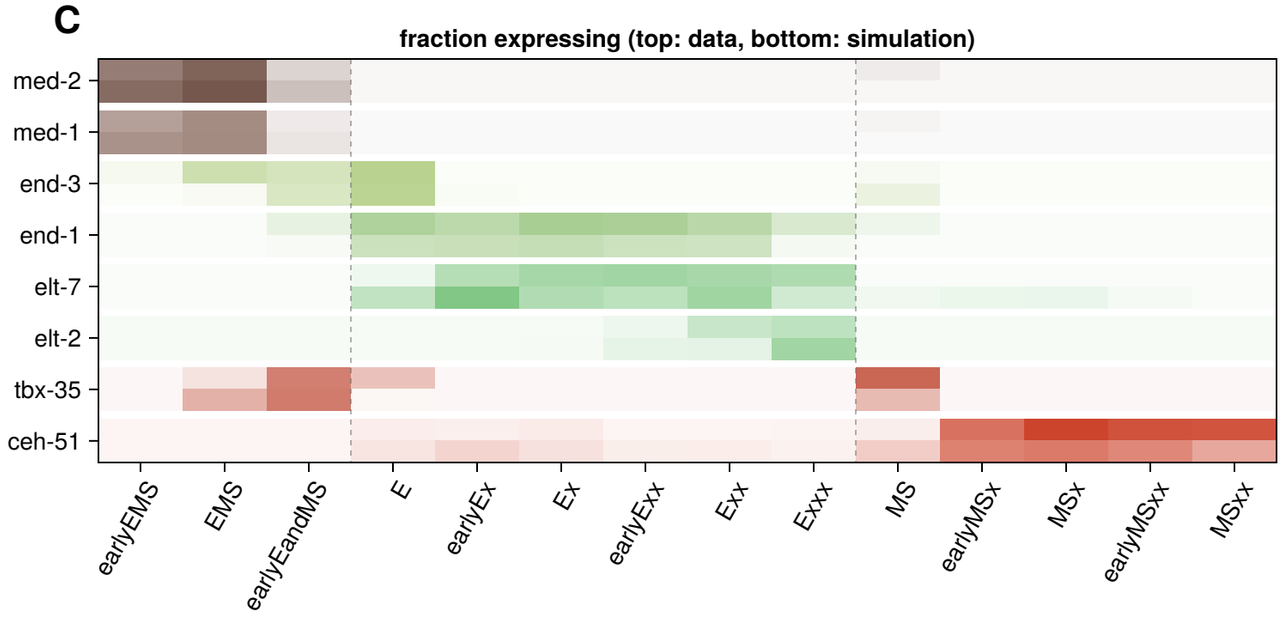
A


snRNA-seq
×20 reps
knockouts
×80 reps
+ initial SKN-1, POP-1, REF-1
earlyEandMS
earlyMSx
earlyMSxx
MS
MSx
MSxx
earlyEMS
EMS
◇ readout · pre-mRNA · 20% capture
E
Ex
Exx
earlyEandMS
earlyEx
earlyExx
Exxx
+ P2 signal
END-1 protein readout
END-1 protein readout
KO(elt-2ⁱ+elt-7ⁱ)
KO(elt-7ⁱ+end-1)
ELT-2 protein readout
0
20
40
60
80
100
120
140
160
180
200
220
240
260
280
300
minutes after EMS cell birth

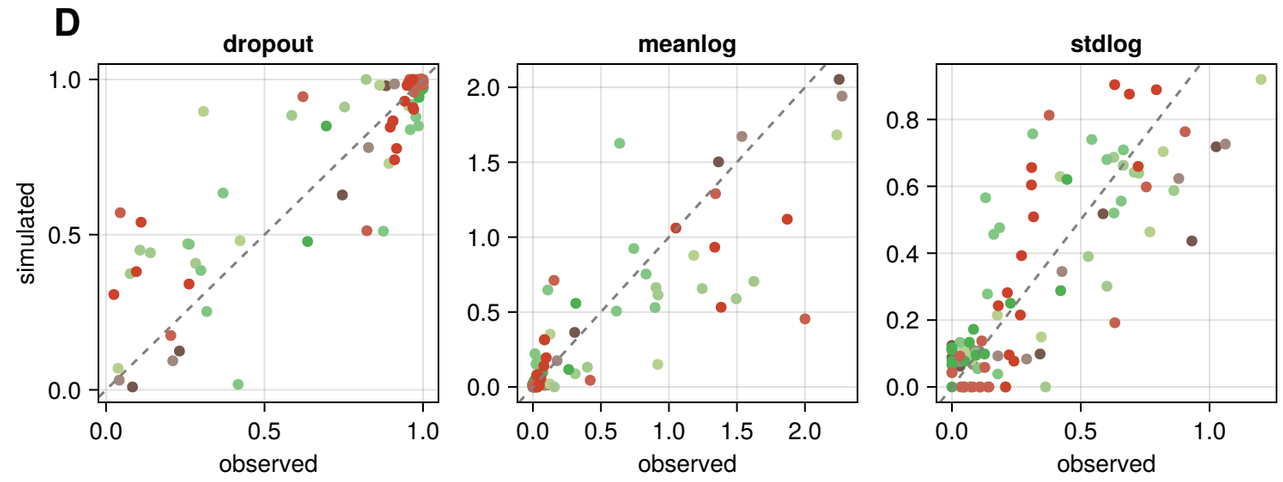
B


P2-signal
pop-1.sys-1_binding
pop-1
POP-1
POP-1*
ref-1.activation
ref-1
REF-1
REF-1*
skn-1
med-2
med-1
tbx-35
end-3
end-1
ceh-51
elt-2
elt-7

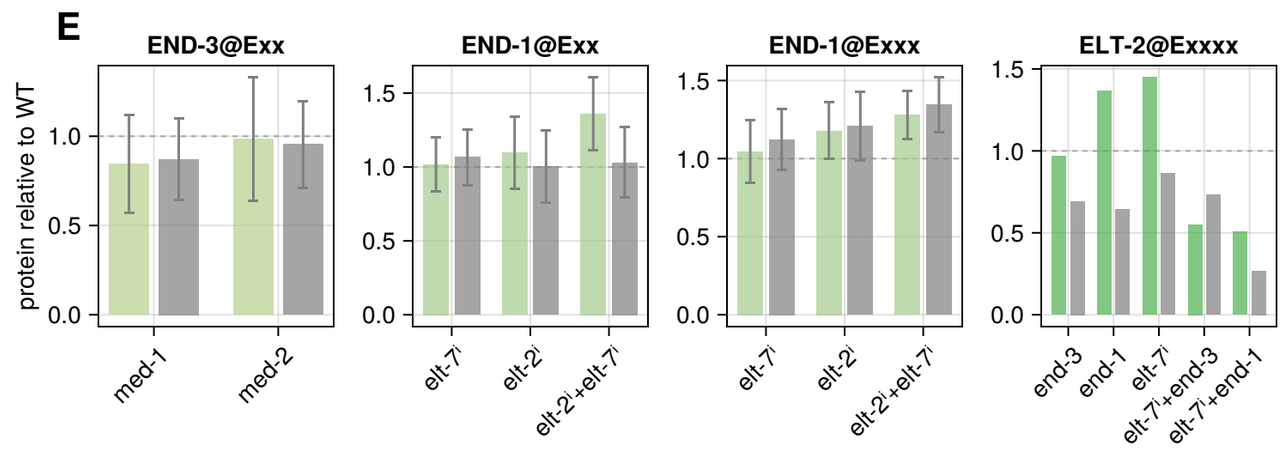
C


fraction expressing (top: data, bottom: simulation)
med-2
med-1
end-3
end-1
elt-7
elt-2
tbx-35
ceh-51
earlyEMS
EMS
earlyEandMS
E
earlyEx
Ex
earlyExx
Exx
Exxx
MS
earlyMSx
MSx
earlyMSxx
MSxx

D


dropout
meanlog
stdlog
simulated
observed
0.0
0.5
1.0
1.5
2.0
0.2
0.4
0.6
0.8

E


END-3@Exx
END-1@Exx
END-1@Exxx
ELT-2@Exxxx
protein relative to WT
0.0
0.5
1.0
1.5
med-1
med-2
elt-7ⁱ
elt-2ⁱ
elt-2ⁱ+elt-7ⁱ
end-3
end-1
elt-7ⁱ+end-3
elt-7ⁱ+end-1